Information Retrieval
Programming Assignment II
Vector Space Model and Ranked Retrieval
All your file names should be prefixed by your group name.
Submission deadline: September 12.

General Guidelines: The project is to be done in a group. ONLY ONE MEMBER PER GROUP SHOULD SUBMIT THE ASSIGNMENT IN MOODLE. Plagiarism will be penalised. Marks will be awarded depending on completeness and correctness. The assignment uses the open-source search engine terrier/lucene. You need to submit your programs as well as a writeup on your implementation details, plan of experiments and results.

Text Collection: You will use the Cranfield text collection. The file cran.tar.gz has all the relevant data.

Problem Description:
The assignment is about using open-source search engines like pyterrier/lucene to build ranked retrieval pipelines. In the assignment you will perform ranked retrieval on the Cranfield text collection and evaluate the retrieval performance on the test queries and the human relevance judgments provided for these queries. You should experiment with different retrieval models and parameter settings. Usual preprocessing steps performed in the previous experiment should be repeated. Parameter and model tuning should be performed to maximise retrieval performance. A report containing a comparative retrieval performance of different parameter settings and models should be prepared. The search and indexing time should also be noted. You may use only sparse vector space models. Don't use advanced probabilistic of dense neural retrieval models.

Submissions: All the program files, result files for the test queries as a text file, and a report as a pdf file. The report may be about 10 pages long. The report should contain:

1. Group details
2. Problem statement
3. Implementation details
4. Plan of experiments
5. Results
6. Discussion of results

The submitted files will be evaluated on a further set of unknown test queries using an evaluation script. The top three groups with the best performance will receive bonus points.

Links to resources:
PyTerrier Documentation and Tutorial:

https://pyterrier-tutorial.github.io/

https://pyterrier.readthedocs.io/en/latest/

https://github.com/terrier-org/pyterrier

Lucene Documentation and Tutorial:

https://lucene.apache.org/core/9_1_0/index.html

https://web.cs.ucla.edu/classes/winter15/cs144/projects/lucene/index.html